SE(3)-WAM: Role-Aware 4D World–Action Modeling for Dynamic Manipulation Supplementary Material

Anonymous CVPR submission

A. Method and Training Details

A.1. Error Decomposition

This subsection derives Eq. 2 and the rotation and context terms that Table S3 assigns to the tasks. It describes how the error of an estimate scales with the lead and the motion; it is not a model of success.

Translation. Write the true target position at the execution time as $\mathbf{p}(\tau^{\mathrm{a}}) = \mathbf{p}(\tau_t) + \mathbf{v}\Delta + \boldsymbol{\rho}$, with the velocity $\mathbf{v}$ at τ_t and the remaining motion $\boldsymbol{\rho}$, and the policy's estimate as $\hat{\mathbf{p}}(\tau^{\mathrm{a}}) = \hat{\mathbf{p}}(\tau_t) + \hat{\mathbf{v}}\hat{\Delta} + \hat{\boldsymbol{\rho}}$. Subtracting and adding $\mathbf{v}\hat{\Delta}$ gives

$$\begin{aligned}\hat{\mathbf{p}}(\tau^{\mathrm{a}}) - \mathbf{p}(\tau^{\mathrm{a}}) = \big[\hat{\mathbf{p}}(\tau_t) - \mathbf{p}(\tau_t)\big] + (\hat{\mathbf{v}} - \mathbf{v})\hat{\Delta} \\ + \mathbf{v}(\hat{\Delta} - \Delta) + (\hat{\boldsymbol{\rho}} - \boldsymbol{\rho}),\end{aligned} \tag{S1}$$

and the triangle inequality yields Eq. 2 with $e_v = \|\hat{\mathbf{v}} - \mathbf{v}\|$ and $e_{\mathrm{dyn}} = \|\hat{\boldsymbol{\rho}} - \boldsymbol{\rho}\|$. The residual $\boldsymbol{\rho}$ is the motion that constant-velocity extrapolation misses. Under a constant acceleration $\mathbf{a}$, $\boldsymbol{\rho} = \frac{1}{2}\mathbf{a}\Delta^2$, which an acceleration estimate from the history reduces; impacts, rolling contact, and friction changes leave a remainder that only a model of the dynamics can reduce.

Rotation and the contact point. For rotations about a common fixed axis, with $R(\tau^{\mathrm{a}}) = \mathrm{Exp}(\boldsymbol{\omega}\Delta + \boldsymbol{\eta})R(\tau_t)$ and its estimate defined likewise, the same steps give

$$\begin{aligned}\theta &= \big\| \mathrm{Log}(\hat{R}(\tau^{\mathrm{a}})R(\tau^{\mathrm{a}})^{\top})^{\vee} \big\| \\ &\le \theta_{\mathrm{loc}} + \hat{\Delta}\, e_\omega + \|\boldsymbol{\omega}\|\, e_\tau + e_\eta;\end{aligned} \tag{S2}$$

general rotations add terms of second order in the angles. A contact point at $\mathbf{c}$ in the object frame, with $\|\mathbf{c}\| = r$, is off by $(\hat{\mathbf{p}} - \mathbf{p}) + (\hat{R}R^{\top} - I)R\mathbf{c}$, and the operator norm of $\hat{R}R^{\top} - I$ is $2\sin(\theta/2)$, so

$$e_{\mathrm{c}} \le \|\hat{\mathbf{p}} - \mathbf{p}\| + 2r\sin(\theta/2) \le \|\hat{\mathbf{p}} - \mathbf{p}\| + r\theta. \tag{S3}$$

The orientation error thus enters the contact scaled by the lever arm, and a grasp or placement that depends on orientation needs the full SE(3) state of the target.

Context. If the executed path keeps every point of the robot and the carried object at least m away from the *predicted* positions of the context points it passes, a collision requires a context error of at least m, so a coarse estimate of the surroundings suffices when $m \gg \varepsilon$. SE(3)-WAM does not enforce this margin, so the no-scene-flow control of Sec. 4.3 tests the effect of coarse context supervision; a flow sampled on a grid can also miss structures thinner than its spacing. A driver, such as a conveyor, a ramp, or a striker, affects the contact only through the motion it imparts to the target, mostly through e_{dyn}, which the SE(3) forecast supervises.

A.2. Input Tokens

Current image. A frozen MAE ViT-B/16 [4] encodes the 224×224 agent-view RGB into 196 patch tokens; we chose it over frozen DINOv3 encoders [11] and a CNN trained with the policy on development scenes (Appendix E.1). Each patch receives a view embedding, a linear embedding of its mean valid metric depth, and a linear embedding of the base-frame position of its back-projected center pixel, computed with the camera intrinsics and extrinsics. These positions also enter 3D rotary position embeddings of the patch queries and keys, and the same center points are the source points of the sparse 3D flow (Appendix A.3). Target, robot, summary, and action tokens receive no rotary embedding. The 196 patch tokens, 8 target-state tokens, and 8 robot tokens form the 212 scene tokens; past images never enter the model.

Target-motion history. The target history covers the last 8 control steps, at the offsets $-350, -300, \ldots, 0$ ms. Each of the 8 target-state tokens is a 25-dimensional descriptor $(\mathbf{p}, \log R, \Delta\mathbf{p}, \log \Delta R, \Delta t, \mathbf{v}, \boldsymbol{\omega}, \mathbf{a}, \boldsymbol{\alpha})$ with dimensions $(3, 3, 3, 3, 1, 3, 3, 3, 3)$. Slots 0–6 describe the seven intervals between consecutive offsets, and slot 7 holds the current absolute pose in the base frame. Velocities divide interval displacements by the elapsed time, and accelerations divide adjacent velocity differences by the spacing of the interval centers. We standardize each dimension with training statistics, zero invalid dimensions, and append three validity flags (pose, interval, and acceleration) before a $28 \to 1024 \to 1024$ GELU MLP with an additive relative-

time embedding. A slot without any valid feature is masked out of attention. On the real robot, we resample the tracker's estimates (Appendix D.2) at the same offsets, linearly in translation and along the geodesic in rotation, without extrapolation and without interpolating across gaps longer than two camera periods.

Robot state and summary tokens. 8 end-effector states (position, rotation vector, and measured gripper opening fraction) at the same offsets form the robot tokens. 16 learned embeddings form the summary tokens. The policy has no language token, and the view, time, depth, and position embeddings modify existing tokens rather than adding slots.

A.3. Forecast Heads and Losses

All forecast heads are light readouts of the final hidden states of the world stream. They are trained jointly with the action stream and skipped at deployment, and no token reads their outputs.

Target. A linear map from the mean final state of the summary tokens gives $\{(\hat{\mathbf{d}}_h, \hat{\boldsymbol{\phi}}_h)\}_{h=1}^{8}$, trained with $\mathcal{L}_P$ of Eq. 4 and $\lambda_R = 0.1$.

Sparse 3D flow. The flow is the cumulative base-frame displacement of the 196 patch center points at the 8 horizons, together with their visibility. A linear readout on each patch token gives its point's displacement and visibility logit at every horizon, trained with $\mathcal{L}_F$ of Eq. 5 and the scale $s = 0.7154605$ m. The visibility loss has weight 1 in simulation, and points that are valid but invisible still supervise their displacement. The points cover the target, the robot, and the surroundings alike.

Auxiliary targets. Linear readouts of the final patch tokens predict three future appearance targets at every horizon: metric depth and moving-surface RGB through 16×16 pixel readouts per patch, and the 1024-dimensional features of a frozen DINOv3 ViT-L/16 teacher [11] on the 14×14 grid. Depth uses squared error on valid pixels, the DINO features squared error after channel-wise layer normalization, and RGB squared error on surfaces that move by more than 1 mm, including the robot. A $1024 \to 256 \to 3072$ MLP with GELU and a final sigmoid reads the mean of the final summary tokens and reconstructs the current target's 32×32 RGB region with an absolute error. The four losses form $\mathcal{L}_{\text{aux}}$, each with weight $\lambda_{\text{aux}} = 0.1$.

A.4. Action Stream and Backbone

Commands. Each chunk has 8 seven-dimensional commands at a control period of 50 ms, each with a base-frame translation increment, a left-relative rotation vector, and a gripper command. Translation and nonzero rotation increments refer to the measured end-effector pose at application. Command j of the chunk for query t is labeled with the demonstration command issued at $\tau_t + j\Delta_a$. The task's command delay d_{task} is added on application (Eq. 1); it does not shift the training indices.

Flow matching. Normalization uses the mean and population standard deviation of the valid issued commands of the training split. Flow matching uses a unit Gaussian in these coordinates with one flow time u per chunk, which the action tokens receive through adaptive layer normalization. Each prediction position draws 8 independent noise–time pairs and averages their losses, which reduces the variance of the action gradient without enlarging the batch. At deployment, we integrate 5 Euler steps, reverse the normalization, decode the rotation vectors, and clip each axis to 0.05 m and 0.5 rad; the gripper follows the sign of its command.

Backbone and mask. Each of the 24 layers applies one attention over all tokens with shared query, key, value, and output projections, followed by separate SiLU feed-forward networks (ratio 4) for the world and the action tokens, for 504.0M backbone parameters. The mask acts on the scene ($\mathcal{S}$), the summary tokens ($\mathcal{M}$), and the actions ($\mathcal{A}$): scene tokens attend to $\mathcal{S}$, summary tokens to $\mathcal{S}$ and $\mathcal{M}$, and action tokens to $\mathcal{M}$ and $\mathcal{A}$. A blocked edge has the logit $-\infty$ in every layer and head, in training and at deployment. Because the world tokens attend neither to the actions nor to the flow time, their states are computed once per query and reused across the Euler steps. Appendix B.3 specifies how the resulting chunks are executed.

A.5. Training on Dynamic-14

Table S1 lists the recipe on Dynamic-14. Each control step of a demonstration anchors one training window, which gives 1,324,374 windows from the 11,200 demonstrations; 350 further episodes form the validation split. A window holds 8 query positions, each with its own causal history and no attention across positions. Each loss is normalized by its count of valid labels over the accumulated global batch. The DINOv3 features of future frames are cached once per episode.

A.6. Pretraining on DROID and Fine-Tuning

Data. The real-robot base trains on DROID [6], recorded at 15 Hz, with the 3D point tracks and optimized camera calibrations that PointWorld released for it [5]. We associate each released clip with its raw DROID episode by exact agreement of the joint sequence, end-effector poses, gripper states, commands, and camera identity, and exclude clips with a non-monotonic capture clock or an ambiguous association. The model sees the first exterior camera with its

Table S1. Training hyperparameters on Dynamic-14.

Item	Value
Backbone	24 layers, width 1024, 16 heads, FFN 4096, 504.0M parameters
Tokens	212 scene, 16 summary, 8 action
Encoder	frozen MAE ViT-B/16, 224×224 input
Optimizer	AdamW, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, weight decay 10^{-4}
Learning rate	2×10^{-4}, 5,174 warm-up updates from 0, cosine decay to 0
Batch, updates	global batch 512 windows, 20 epochs (51,740 updates)
Loss weights	$\lambda_P = \lambda_F = 1$, $\lambda_R = 0.1$, $\lambda_{\text{aux}} = 0.1$ per target
Precision	trainable weights FP32, frozen encoder BF16, gradient clipping at 0.1
Action noise	8 noise–time pairs per query position
Sampling	5 Euler steps for the action
Inference	FP32, world states cached within a query
Hardware	[N] RTX PRO 6000 GPUs, [X] hours per run

depth and PointWorld's calibration. Of 42,935 associated episodes, 42,087 yield 598,889 anchor windows, 538,740 from 37,874 training episodes and 60,149 from 4,213 validation episodes, with no shared episode. Each anchor expands into 8 consecutive query positions with their own causal histories; the anchor carries the flow and DINOv3 labels, and the other positions carry action labels only. All times are capture timestamps: the robot history uses the 8 most recent states at or before the query time, and the flow and DINOv3 targets keep PointWorld's record times rather than being interpolated to 50 ms horizons. Robot states and commands are expressed in the base frame after the calibrated flange-to-TCP transform. We keep the 96.88% of action slots at which the collector issued a command, without a filter on magnitude or speed, so that the base is not biased toward slow motion.

What the base learns. DROID provides no object poses, so the base has no target-state tokens and no pose head. Its inputs are the current RGB-D frame and the robot history (204 scene tokens), with unavailable depth masked, and its objectives are the action loss (weight 1), the sparse 3D flow (weight 1), and the future DINOv3 features (weight 0.1). PointWorld's tracks record only visible positions, so the flow labels contain no negative visibility examples, and the visibility loss is off. Most objects in DROID move only when the robot moves them, so the base mainly learns metric geometry, the motion caused by the robot's own actions, and its coupling with the actions; the exogenous motion of our tasks and the target pose enter during fine-tuning.

Flow labels. Each patch of the 14×14 grid follows one PointWorld track over the window. Among the tracks that start inside the patch, are initially visible, and have a valid positive depth, we prefer the one with the largest image motion whose 3D displacement agrees with its image motion, and otherwise take the track nearest to the patch center. This rule covers moving objects that a fixed center point would miss: on 18 segments with visible object motion, it placed a dynamic track inside the object in 13, against 5 for the center rule. The label is the track's displacement from its initial position, and patches without a valid track are invalid; 47.05% of the point–horizon slots carry a label.

Recipe. The base uses 48 layers at width 1024 (1.02B trainable parameters) and trains for 20 epochs (21,060 updates, 2,106 warm-up updates) at a global batch size of 512 on 64 GPUs, with the optimizer and precision of Table S1.

Fine-tuning. Each real-robot policy is initialized from the base and fine-tuned on the demonstrations of its task. The target-state tokens, the pose head, and the appearance heads that the base lacks start from fresh initialization, and the policy receives poses from the tracker of Appendix D.2. No real-robot policy is initialized from simulation weights, and the base sees no simulation data.

B. Dynamic-14 and Evaluation Protocol

B.1. Tasks, Roles, and Success Conditions

We built Dynamic-14 on MuJoCo and robosuite [12, 18]. It contains fourteen tasks in five families (Fig. 4). Table S2 lists their success conditions and Table S4 their scenario ranges. We will release the tasks, the accepted scenarios, the success checks, and the demonstrations.

Roles. Table S3 assigns the bodies of each task to the target and context roles of Sec. 3.1. A moving receptacle is therefore a target, since its pose decides containment. The wire of the passage tasks plays both roles: its pose at the entry is a target, because the wand must be aligned within 30 mm there, and its geometry along the loop is context, because any contact fails. Conveyors, turntables, ramps, and strikers are drivers in the context. The last column lists the error terms of Eq. 2 that each task exercises (Appendix A.1). The two task groups of Fig. 5 follow from this table and were fixed before the ablations: the rotation-sensitive group contains the tasks whose terms include $r\theta$ (`grasp`, `t1_so3`, `t2_se3`, `t3_phase`, `t3_full`, `t4_can`, `t5_replan`), and the context-active group contains the tasks with an obstacle clearance or a moving context (`t1_belt`, `t1_occ`, `t3_phase`, `t3_full`, and the four capture tasks and `t5_replan`).

Table S2. Success conditions of the fourteen Dynamic-14 tasks, with the thresholds of our task implementations. A task must pass its preceding stages and avoid terminal failures. All tasks run at 20 Hz with a 2 ms physics step; durations below refer to physical time. COM denotes center of mass.

Task	Ordered success conditions	Limit
grasp; t1_belt, t1_xyz, t1_so3, t1_occ	Approach within 0.10 m; bilateral fingerpad contact for 0.05 s; collision-geometry bottom at least 0.05 m above support; retain contact and clearance for 0.50 s. T1 adds conveyor drive, translation, rotation, or a physical occluder. SO3 has no additional success-angle threshold; unsafe occluder contact terminates OCC.	160 steps 8 s
t2_trans	Approach within 0.10 m. Entire object collision geometry inside the receptacle: planar clearance and floor gap both ≥ -0.002 m, top no higher than the opening. Release requires an opening command < -0.5 and no fingerpad contact. Relative linear/angular speeds ≤ 0.08 m/s and 0.8 rad/s for 0.10 s, followed by a 0.50 s stable dwell.	200 10 s
t2_se3	Approach within 0.05 m and 0.12 rad. Full T-shaped containment: planar clearance ≥ 0.002 m, floor gap in $[-0.002, 0.002]$ m, tilt ≤ 0.05 rad, top below opening. After release, relative speeds ≤ 0.02 m/s and 0.04 rad/s for 0.10 s, then a 0.50 s dwell. The 0.12 rad gate applies to approach.	200 10 s
t3_phase	The episode starts with the wand already grasped. Phase alignment: distance ≤ 0.090 m and loop-direction error ≤ 0.20 rad. Enter within 0.030 m at progress ≤ 0.15; reach historical maximum progress ≥ 0.60, then ≥ 0.85 and endpoint distance ≤ 0.055 m.	420 21 s
t3_full	Acquire the wand with bilateral contact for 0.05 s, then complete the phase stages above. In a later sample, retain current progress ≥ 0.85 and move at least 0.045 m from both path and endpoint. Finally open (< -0.5) with no fingerpad contact. Releasing before exit fails.	640 32 s
t4_sphere	Approach within 0.10 m; first qualifying contact after ramp exit; any tray contact before success fails. Bilateral hold for nominal 25 ms (12 physics samples, 24 ms). Raise COM by 0.05 m from its height when the bilateral-hold criterion is first met, and retain contact and height for 0.50 s.	120 6 s
t4_sphere_tabletop; t4_slider	Approach within 0.10 m; contact after ramp exit, while supported, in the capture region, and moving at planar speed ≥ 0.08 m/s. Bilateral hold for 0.05 s, COM lift ≥ 0.05 m from the first attainment of that hold criterion, then contact and height for 0.50 s.	120 6 s
t4_can	Same supported-capture sequence. Before first fingerpad contact, a can-axis absolute dot product with vertical > 0.45 fails as unstable tipping. The configuration does not require a transverse closing axis.	120 6 s
t5_replan	Exactly one selected-side impact group completes before robot contact. At 30 ms after the physical impact-end timestamp, branch-signed lateral speed ≥ 0.035 m/s and yaw rate ≥ 0.35 rad/s. Approach within 0.10 m; bilateral hold with grasp-axis error $\leq 18^\circ$ for 0.05 s; COM lift ≥ 0.05 m from the first attainment of that hold criterion and stable hold for 0.50 s. Impact-group lag is 12 ms and maximum contact is 0.10 s.	120 6 s

Terminal events. We check success at every 2 ms physics step, after the ordered requirements of Table S2, and an interrupted continuous condition resets its counter. Terminal events fall into five categories. *Invalid* events are corrupted simulator states. *Boundary* events are a workspace or downstream exit, a drop, or an expired goal window. *Safety* events are unsafe contact, an excessive rim impulse, wire contact, a release impulse, early robot contact, unsupported tray contact, or a striker trap. *Task violations* are a release outside the goal, a path deviation or backtracking, a premature release, a deflection without capture, tipping, a missing or extra impact, or an insufficient change of outgoing motion. The last category is the *timeout*. Each family evaluates its events in a fixed order per physics step and keeps the first terminal event.

B.2. Scenarios, Demonstrations, and Splits

We sample continuous ranges uniformly and discrete choices with equal probability, and then apply physical-feasibility checks. Evaluation replays the stored accepted scenarios. Harmonic translation follows $\mathbf{p}(t) = \mathbf{c} + (0, v_y, 0)t + (A_x \sin(2\pi f_x t + \phi), 0, 0)$ with $\phi \sim \mathcal{U}[-\pi, \pi]$. Capture targets start at rest and move under gravity and friction on ramps of $[5, 15]^\circ$. We accept an occluded scene only if the target is visible for at least eight initial frames, fully occluded for 0.5–3.0 s, and visible again for at least three frames.

The training pool contains 11,200 demonstrations, 800 per task, generated by experts with privileged simulator state; the learned policies receive only the inputs of Sec. 3. We record the demonstrations at 20 Hz with camera images, depth, robot states, issued commands, and simulator ground truth for target poses and scene geometry, from which we

Table S3. **Roles and tolerances of the Dynamic-14 tasks.** The target is the body whose pose error enters a contact or containment tolerance ε. The context comprises obstacles, avoided with clearance m, and drivers that move the target. Speeds follow the sampling ranges of Table S4 (capture tasks: at the ramp exit). Terms: error terms that a task exercises (Eq. 2 and Appendix A.1). v: moving target. $r\theta$: orientation enters the contact. e_{dyn}: non-constant motion (harmonic, gravity, friction, impact). m: a context body must be avoided. Occ.: history gaps. Delay: command delay built into the task.

Task	Code	Target	Context	ε (mm)	$\|\mathbf{v}\|$ (m/s)	m (mm)	Terms
Harmonic grasp	grasp	block	table	[X]	0.03–0.20	–	v, $r\theta$, e_{dyn}
Conveyor grasp	t1_belt	part	belt	[X]	0.04–0.10	–	v
Translating grasp	t1_xyz	part	table	[X]	0.04–0.15	–	v, e_{dyn}
Rotating grasp	t1_so3	part ($\omega \leq 1.2$ rad/s)	table	[X]	0.04–0.15	–	v, $r\theta$, e_{dyn}
Occluded grasp	t1_occ	part	occluder canopy	[X]	0.04–0.10	[X]	v, m, occ.
Moving receptacle	t2_trans	receptacle	table	[X]	0.03–0.06	–	v
Rotating receptacle	t2_se3	T-shaped receptacle	table	[X]	0.03–0.06	–	v, $r\theta$
Wire passage	t3_phase	wire loop (entry)	rotating wire	30	[X]	swept	v, $r\theta$, m
Wire passage, full	t3_full	wire loop (entry)	rotating wire	30	[X]	swept	v, $r\theta$, m
Ball capture	t4_sphere	ball	tray, ramp	[X]	[0.4–0.7]	[X]	v, e_{dyn}, m
Ball capture, table	t4_sphere_tabletop	ball	table, ramp	[X]	[0.4–0.7]	–	v, e_{dyn}
Slider capture	t4_slider	slider	table, ramp	[X]	[0.4–0.8]	–	v, e_{dyn}
Can capture	t4_can	can	table, ramp	[X]	[0.4–0.7]	–	v, $r\theta$, e_{dyn}
Post-impact grasp	t5_replan	part	striker	[X]	0.12–0.14	[X]	v, $r\theta$, e_{dyn}, m, delay

compute the future labels, their validity masks, and the command labels.

We select configurations on 1,400 development scenes (100 per task). All Dynamic-14 results use a separate held-out set of 1,400 scenes generated with the same ranges and new source seeds that appear in no training, validation, or development episode; the first 100 feasible scenes in increasing seed order form each task's set.

B.3. Execution Protocol and Latency Measurement

Executor. Every method runs under the executor of Sec. 3.4 on a 50 ms control clock, with queries starting on its period boundaries. The total latency L_t of query t is the time that physics advances while it runs: its inference time under the protocol plus any added delay. Command j of its chunk executes at $\tau^{\text{a}}_{t,j}$ of Eq. 1, with $d_{\text{task}} = 50$ ms for the post-impact grasp and zero otherwise. Its commands $j < k_t$, with $k_t = \lceil L_t/\Delta_a \rceil$, have expired and are discarded, so query t executes commands $j = k_t, \ldots, \min(8, k_t + k_{t+1}) - 1$, one per control period. When $k_t + k_{t+1} > 8$, e.g., for a constant total latency above 200 ms, a chunk runs out before the next one arrives, and if $k_t \geq 8$, all its commands have expired. Before the first chunk of an episode arrives, and whenever a chunk runs out, the controller holds the current arm target and gripper command while physics advances; no command is repeated. The reported time-to-action is the measured latency before rounding to $k_t\Delta_a$.

Real-time protocol. Unless noted otherwise, physics advances during inference by the method's native latency, drawn per query from its latency distribution measured at batch size 1 on one RTX 5090. The native-latency ACT/DP rollouts run on RTX 5090; ACT paused and added-latency rollouts and the planned remaining evaluations use RTX PRO 6000. Their own compute time does not enter the episode. Table S7 reports three alternatives. With *paused* physics, the executor is unchanged and the simulation waits for inference; an added delay still advances physics. Without added delay, $L_t = 0$, and command 0 executes at capture. With *equal latency*, every method is charged the p95 latency of SE(3)-WAM regardless of its own speed. With *original executors*, each baseline keeps the execution scheme of its paper, e.g., a longer chunk before replanning.

Added latency. An added latency of 0, 50, 100, 150, or 200 ms is inserted after inference on top of the protocol's latency, and physics advances during it.

Latency measurement. We time all methods on one RTX 5090 (32 GB), the GPU of our robot, with exclusive use and batch size 1. These measurements form the latency distributions that the simulated rollouts replay independently of the rollout hardware. For each method, each of three rounds uses thirty warm-up calls and 300 measured calls over three observation sequences of eight frames, with CUDA synchronization at call boundaries and a rotated method order. Each method uses the precision of its release; SE(3)-WAM uses FP32. The time-to-action includes encoding the image and generating the first command, and on the robot also tracking the newest frame. For SE(3)-WAM, we time the forecast heads separately, although deployment skips them. We distinguish three times. The *time-to-action* is the measured wall-clock time from the arrival of an observation to the availability of its first command, without rounding; the

Table S4. Scenario proposal ranges. Length, linear velocity, angle, and angular velocity use m, m/s, rad, and rad/s unless marked otherwise. Positions are in world coordinates except the stated ramp-local offsets. Continuous intervals are uniform proposals; discrete choices are equiprobable before feasibility checks. Evaluation replays the stored accepted scenarios, whose distribution need not remain uniform. Harmonic centers are not initial positions.

Task	Initial-state proposal	Motion or event parameters
grasp	Harmonic center $c_x \in [-.08, .08]$, $c_y \in [-.26, -.19]$, $z = .822$; yaw $[-1.1, 1.1]$.	$v_y \in [.025, .125]$, $A_x \in [.015, .035]$, $f_x \in [.3, .7]$ Hz; $\omega_z \in [-1.2, 1.2]$.
t1_belt	Target $x \in [-.08, .08]$, $y \in [-.26, -.18]$, $z = .822$; identity orientation, zero initial twist.	Belt $v_y \in [.04, .10]$.
t1_occ	Same target proposal as Belt; canopy at $(.322, -.29, 1.20)$.	Same belt range; clearance parameter .20; physical occlusion.
t1_xyz	Harmonic center $c_x \in [-.025, .025]$, $c_y \in [-.24, -.20]$, $z = .818$; yaw $[-.35, .35]$.	$v_y \in [.04, .09]$, $A_x \in [0, .03]$, $f_x \in [.25, .65]$ Hz; $\omega_z = 0$.
t1_so3	Same as XYZ.	Same translation; $\omega_z \in [-1.2, -.4] \cup [.4, 1.2]$, equal sign probabilities.
t2_trans	Goal $x \in [-.13, -.08]$, $y \in [-.18, -.12]$, $z = .825$; fixed orientation.	Goal $v_y \in [.03, .06]$; no rotation.
t2_se3	Same goal x, y range, $z = .804$; yaw $[-.3, .3]$.	Goal $v_y \in [.03, .06]$; $\omega_z \in [-.25, -.08] \cup [.08, .25]$, equal sign probabilities.
t3_phase	Wand $x, y \in [-.03, .03]$, $z = .870$, yaw $[0, 2\pi]$; wire phase $[-.25, .35]$.	Wire center $(.10, -.02, .94)$; world-X rotation $\omega_x = -\{2.5, 3, \ldots, 7.5\}^\circ$/s (11 equal-probability values).
t3_full	Wand at $(-.05, -.02, 1.04)$ with identity orientation; same wire phase range.	Same wire rotation; cradle and release requirement enabled.
t4_sphere	Ramp-local offsets: lateral $[-.015, .015]$, longitudinal/normal $(-.145, .030)$; lip $(.06, -.02, 1.10)$.	Gravity-driven from rest; gate release at tick 0; slope $[5, 15]^\circ$; target/ramp sliding friction $[.7, .9]$; unsupported capture.
t4_sphere_tabletop	Lateral $[-.015, .015]$, longitudinal/normal $(-.124, .034)$; lip $(0, -.06, .808)$.	Same slope and release; target/ramp friction $[.7, .9]$, table friction .35.
t4_slider	Lateral $[-.015, .015]$, longitudinal/normal $(-.129, .024)$; supported ramp.	Same slope and release; target/ramp friction $[.025, .045]$, table friction $[.012, .018]$.
t4_can	Lateral $[-.010, .010]$, ramp-axis yaw $[-4, 4]^\circ$; longitudinal position follows gate geometry and oriented can extent.	Same slope and release; target/ramp friction $[.25, .45]$, table friction $[.09, .15]$.
t5_replan	$x \in [-.005, .005]$, $y = -.29$, $z = .812$; yaw $[-.08, .08]$; zero initial twist.	Belt $v_y \in [.12, .14]$; equal-probability left/right striker triggered at $y = -.13$; stroke .098, pulse .35 s, actuator scale $[1, 1.1]$, force cap 4 N; command delay 50 ms.

handover time $\ell_t = k_t \Delta_a$ rounds the total latency up to the control period; and the *action age* is the execution time of a command minus the capture time. On the robot, traces record capture, availability, preprocessing, the start and end of inference, queue entry, command issue, and execution. A query misses its deadline when its time-to-action exceeds the 50 ms control period, so that the next period executes a remaining command or holds. Appendix C.2 reports the measured values.

B.4. Baselines and Ablation Controls

Common conditions. Table S5 summarizes the adaptations. Each baseline keeps its official architecture, action chunk, and initialization. ACT and Diffusion Policy train from scratch, $\pi_{0.5}$ is fine-tuned from `pi05_base`, and the other methods start from their released weights. All of them train on the same 11,200 demonstrations, whose commands we map to each action space through the same Cartesian controller. The seven trained baseline endpoints use agent-view RGB only, as does SE(3)-WAM. The native missing-view paths are retained: $\pi_{0.5}$ masks its two zero wrist slots; DreamVLA uses a preprocessed black wrist image with wrist auxiliary supervision masked; OpenWAM-α and DynamicWAM fill their missing wrist regions with black. Each baseline follows its official update budget, trained once with seed 0; the input and action adaptations are stated here and in Table S5. For DreamVLA, we ran both its original 40-epoch schedule and a 20-epoch cosine schedule and report the one that was better on development scenes.

Table S5. **Baseline adaptations.** All methods train on the same 11,200 demonstrations with the inputs listed below and run under the shared executor of Appendix B.3. Updates × batch follows the official recipe. Exec.: the execution scheme of the original paper (commands executed per chunk before replanning), used only for the original-executor control of Table S7. Params: trainable parameters, frozen encoders excluded. Seeds: training seeds.

Method	Initialization	Inputs	Updates × batch	Exec.	Params	Seeds
ACT [17]	scratch	current agent RGB, robot state	50,000 × 32	temporal agg.	[X]	1
Diffusion Policy [1]	scratch	two causal agent RGB frames, robot state	50,000 × 32	8	[X]	1
$\pi_{0.5}$ [9]	`pi05_base`	current agent RGB, robot state, task text; wrist slots masked	30,000 × 256	10	[X]	1
$\pi_{0.5}$ + SE(3)	`pi05_base`	planned: $\pi_{0.5}$ + SE(3) tokens; camera inputs pending	30,000 × 256	10	[X]	1
DynamicVLA [15]	released DOM ckpt.	frames $[-2, 0]$, robot state, task text	50,000 × 44	streaming	[X]	1
PUMA [2]	official checkpoint	RGB, optical-flow history, robot state, task text	[X]×[X]	[X]	[X]	1
DreamVLA [16]	released components	seven causal agent RGB frames, robot state, task text; black wrist input	48,220 × 512	1	[X]	1
OpenWAM-α [13]	official foundation	current agent RGB, joints, gripper, task text; missing views black	34,490 × 192	32	[X]	1
DynamicWAM [8]	official initialization	RGB, flow history, kinematic descriptors	40,000 × 128	16	[X]	1
DynamicWAM + SE(3)	official initialization	planned: DynamicWAM + SE(3) tokens; camera inputs pending	40,000 × 128	16	[X]	1
SE(3)-WAM	frozen MAE encoder	current agent RGB-D, SE(3) and robot histories	51,740 × 512	1	523.8M	3

Inputs. PUMA [2] receives its historical optical flow, computed on the rendered frames. DreamVLA receives seven causal RGB frames, the end-effector position and orientation, the finger positions, and a fixed task text, and it samples with ten DDIM steps and classifier-free guidance of 1.5. OpenWAM-α receives the current RGB views, joint values, gripper opening, and task text, and it generates video and actions jointly with video decoding disabled. DynamicWAM receives its flow history and kinematic descriptors.

SE(3)-augmented baselines. $\pi_{0.5}$+SE(3) and DynamicWAM+SE(3) receive the eight target-state tokens of Appendix A.2 from the same pose source, with the same validity flags and masking. A two-layer MLP maps the tokens to the model's width. For $\pi_{0.5}$, we append them to the prefix after the image tokens, where the action expert attends to them; for DynamicWAM, they join its kinematic descriptors. The output layer of the MLP is zero-initialized, and training follows the base model's recipe.

Ablation controls. The five controls of Sec. 4.3 change one component each of the configuration of Table S1 and keep the data, the recipe, the update budget, the executor, and the training seeds 0, 1, and 2. *No target history* removes slots 0–6 of the target-state tokens, which carry the past poses with their velocity and acceleration features, and keeps slot 7 with the current pose (Appendix A.2). *No future supervision* removes $\mathcal{L}_P$, $\mathcal{L}_F$, and the three future appearance losses with their heads, and keeps the reconstruction of the current target region. *No scene-flow supervision* removes $\mathcal{L}_F$ and the flow readout, and *no dedicated SE(3)* removes $\mathcal{L}_P$ and the pose head; both keep the target-state tokens and all appearance targets. The *uniform-fidelity forecast* replaces $\mathcal{L}_P$ and $\mathcal{L}_F$ by [a forecast of future latents of a frozen video VAE of the whole frame at the same horizons, its readout, and its loss weight]. Since every forecast head is skipped at deployment, only the no-history control changes the inputs at inference.

B.5. Metrics and Statistics

The task-balanced success rate is

$$\mathrm{SR}(m) = \frac{100}{14} \sum_{k=1}^{14} \frac{1}{S_m n_k} \sum_{s=1}^{S_m} \sum_{i=1}^{n_k} y_{mski}, \qquad \text{(S4)}$$

for binary outcomes y_{mski} of method m, trained with seed s, on scene i of task k, with $S_m = 3$ training seeds for our variants and $S_m = 1$ for the baselines. The *position error at first contact* e_p is the distance of the end effector from the expert's contact frame on the target at the first contact. The two placement tasks start with the object held, so their contact is the first contact of the held object with the receptacle, and their contact frame is the end-effector pose implied by the expert's placement target. The phase-only passage (`t3_phase`) succeeds without any contact, so e_p averages over the other 13 tasks; we report it on episodes that make contact, and failures without contact stay in the SR denominator.

Training seeds and intervals. The full model and the five ablation controls use training seeds 0, 1, and 2, and each baseline uses seed 0. On development scenes, three training seeds of the same configuration differed by up to [7.4] points in SR, mostly on the wire passage, whose success ranged from [0] to [71]% across seeds. We report each training seed and the cross-seed mean and standard deviation

(Appendix C.4); for the ablations, we also repeat the comparison without the phase-only wire passage (`t3_phase`). For paired 95% intervals, we average the seeds per scene and bootstrap held-out scenes within each task, jointly across methods. These intervals quantify scene uncertainty for the trained weights, and we treat differences whose interval lies within ± 2 points as ties.

C. Additional Simulation Results

C.1. Per-Task Results

Table S6 breaks the held-out Dynamic-14 results of Table 1 down by task for every method, with the paired interval of SE(3)-WAM against the strongest other method on each task; its last two rows give the position error at first contact e_p and the number of episodes with a contact (Appendix B.5). With its native 100-step DDPM sampler, every chunk of Diffusion Policy has expired when it arrives, so none of its commands is executed. [Tasks on which every method fails or a baseline outperforms SE(3)-WAM, and the dominant failure of SE(3)-WAM.]

C.2. Evaluation Protocols and Latency

Table S7 evaluates every method under three further protocols. Paused physics shows how the relative performance changes when the cost of computation is ignored, equal latency compares all methods at the p95 latency of SE(3)-WAM, and the original executors check whether the shared executor removes a capability that a baseline relies on. With paused physics, the difference between SE(3)-WAM and [name] is [$\pm$X.X] points, and under equal latency it is [$\pm$X.X] points. The original executors change the SR of the baselines by at most [X.X] points. Table S8 gives the success at every added latency; its rows for SE(3)-WAM and the strongest baseline are the data of Fig. 1 (right).

Table S7. **Protocol controls** on held-out Dynamic-14 scenes (SR, %). Paused: shared executor, with physics stopped during inference. Equal latency: shared executor, with every method charged the p95 latency of SE(3)-WAM. Original executor: real time with each method's own execution scheme. Real time: the protocol of Table 1. Late: queries in real time whose time-to-action exceeds the 50 ms control period (%). Age: p95 action age in real time (ms); –: no command executed.

Method	Paused	Equal latency	Original executor	Real time	Late (%)	Age p95
ACT	40.9	—	—	42.0	0.0	50
Diffusion Policy	—	—	—	0.0	100.0	–
$\pi_{0.5}$	—	—	—	—	—	—
$\pi_{0.5}$ + SE(3)	—	—	—	—	—	—
DynamicVLA	—	—	—	—	—	—
PUMA	—	—	—	—	—	—
DreamVLA	—	—	—	—	—	—
OpenWAM-α	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—
DynamicWAM + SE(3)	—	—	—	—	—	—
SE(3)-WAM	—	—	–	—	—	—

Table S8. **Success versus added latency** on held-out Dynamic-14 scenes (SR in %, real-time protocol). The 0 and 150 ms columns equal the Native and +150 ms columns of Table 1.

	Added latency (ms)				
Method	0	50	100	150	200
ACT	42.0	—	—	24.1	—
Diffusion Policy	0.0	—	—	—	—
$\pi_{0.5}$	—	—	—	—	—
$\pi_{0.5}$ + SE(3)	—	—	—	—	—
DynamicVLA	—	—	—	—	—
PUMA	—	—	—	—	—
DreamVLA	—	—	—	—	—
OpenWAM-α	—	—	—	—	—
DynamicWAM	—	—	—	—	—
DynamicWAM + SE(3)	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—

Timing. Table S9 breaks the time-to-action of SE(3)-WAM on one RTX 5090 down by stage, measured as in Appendix B.3; Table 1 gives the p95 latency of every method and Table S5 the parameter counts. On the robot, we report the p50, p95, and maximum action age and the fraction of all queries of the test trials that miss the 50 ms deadline, together with the peak memory.

Table S6. **Per-task held-out success** (%) on Dynamic-14 for every method of Table 1, mean of three training seeds for our variants and one for baselines, real-time protocol. DP: Diffusion Policy. DVLA: DynamicVLA. DrVLA: DreamVLA. OWAM: OpenWAM-α. DWAM: DynamicWAM. +: with our SE(3) history tokens. σ: SE(3)-WAM with nominal synthetic correlated pose noise (Appendix C.5). Δ: SE(3)-WAM minus the strongest other method on that task, with its 95% paired interval. The last two rows give the position error at first contact e_p (mm), averaged over the 13 tasks with a contact, and the number n of episodes with a contact (Appendix B.5); –: no 13-task mean. Task codes are listed in Table S3.

Task	ACT	DP	$\pi_{0.5}$	$\pi_{0.5}$+	DVLA	PUMA	DrVLA	OWAM	DWAM	DWAM+	σ	**Ours**	Δ [95% CI]
Harmonic grasp	48.0	0.0	—	—	—	—	—	—	—	—	—	—	—
Conveyor grasp	46.0	0.0	—	—	—	—	—	—	—	—	—	—	—
Translating grasp	18.0	0.0	—	—	—	—	—	—	—	—	—	—	—
Rotating grasp	21.0	0.0	—	—	—	—	—	—	—	—	—	—	—
Occluded grasp	32.0	0.0	—	—	—	—	—	—	—	—	—	—	—
Moving receptacle	26.0	0.0	—	—	—	—	—	—	—	—	—	—	—
Rotating receptacle	0.0	0.0	—	—	—	—	—	—	—	—	—	—	—
Wire passage	99.0	0.0	—	—	—	—	—	—	—	—	—	—	—
Wire passage, full	0.0	0.0	—	—	—	—	—	—	—	—	—	—	—
Ball capture	36.0	0.0	—	—	—	—	—	—	—	—	—	—	—
Ball capture, table	71.0	0.0	—	—	—	—	—	—	—	—	—	—	—
Slider capture	57.0	0.0	—	—	—	—	—	—	—	—	—	—	—
Can capture	90.0	0.0	—	—	—	—	—	—	—	—	—	—	—
Post-impact grasp	44.0	0.0	—	—	—	—	—	—	—	—	—	—	—
Task-balanced	42.0	0.0	—	—	—	—	—	—	—	—	—	—	—
e_p (mm)	32.2	–	—	—	—	—	—	—	—	—	—	—	–
n	1109	6	—	—	—	—	—	—	—	—	—	—	–

Table S9. **Latency breakdown** of one query of SE(3)-WAM on one RTX 5090 at batch size 1 (ms, p50 / p95), and action age on the robot.

Stage	p50	p95
Pose tracking (per frame, robot)	—	—
Image encoding	—	—
World stream (once per query)	—	—
Action stream (5 Euler steps)	—	—
Time-to-action (total)	—	—
Forecast heads (training only, for reference)	—	—
Action age at execution (robot)	—	—
Queries over 50 ms (robot, %)	—	

C.3. Results on DOMINO

On DOMINO [2], we compare SE(3)-WAM, $\pi_{0.5}$, DynamicWAM, and PUMA on 24 single-arm Franka tasks of its clean level-1 setting. All methods use the benchmark's demonstrations, scenes, and native evaluator, one training seed, and 50 episodes per task and method. We report SR and the manipulation score (MS), which scales route completion by penalties for leaving the workspace or view and for collisions (Table 2). SE(3)-WAM uses domain-specific normalization and reads the simulator pose of the target named by the instruction; slots without a valid pose are masked as in Appendix A.2. The RGB-based baselines do not receive this pose. [Summarize the DOMINO results.]

C.4. Final Ablations and Seed Stability

Table S10 reports the five controls of Sec. 4.3 (Appendix B.4) for each training seed, with the cross-seed mean and standard deviation and the paired difference from the full model, with native latency and with 150 ms of added latency. The last two columns repeat the paired differences without the phase-only wire passage (`t3_phase`), whose success varied most across training seeds (Appendix B.5). The two task groups of Fig. 5 are listed in Appendix B.1. [Whether any conclusion of Sec. 4.3 changes for an individual seed or without `t3_phase`.]

C.5. Robustness to Pose Noise

With frozen weights, we evaluate three synthetic perturbations of the target pose: nominal temporally correlated noise, twice its amplitude, and nominal noise with contiguous missing observations. The nominal process combines a fixed per-episode bias and a stationary correlated component, each with per-axis standard deviations of 5 mm and 1°, with a correlation time of 200 ms. Dropouts last 200 ms, separated by geometrically distributed valid gaps with a mean of 3.8 s, giving a long-run missing fraction of 5%. The perturbation applies to the current and past estimates before the history features are computed.

Fig. S1 evaluates the final model under these perturbations, on the same held-out scenes at zero added latency and

Table S10. **Final ablations by training seed** on held-out Dynamic-14 scenes (Appendix C.4). S0–S2: SR (%) of training seeds 0–2; mean ± s.d. over seeds; Δ: control minus SE(3)-WAM (points) with its paired 95% interval. Last two columns: Δ without the phase-only wire passage (`t3_phase`), 13 tasks.

	Native					+150 ms					Δ w/o `t3_phase`	
Variant	S0	S1	S2	Mean ± s.d.	Δ [95%]	S0	S1	S2	Mean ± s.d.	Δ [95%]	Native	+150 ms
SE(3)-WAM *	—	—	—	—	–	—	—	—	—	–	–	–
No target history	—	—	—	—	—	—	—	—	—	—	—	—
No future supervision	—	—	—	—	—	—	—	—	—	—	—	—
No scene-flow supervision	—	—	—	—	—	—	—	—	—	—	—	—
No dedicated SE(3)	—	—	—	—	—	—	—	—	—	—	—	—
Uniform-fidelity forecast	—	—	—	—	—	—	—	—	—	—	—	—

with all three training seeds. Relative to clean poses, SR changes by [X.X], [X.X], and [X.X] points under nominal correlated noise, twice its amplitude, and missing observations.

Figure placeholder: Fig. S1 — pose noise
x: clean poses, nominal correlated noise, twice its amplitude, and nominal noise with missing observations; y: SR of SE(3)-WAM per task family, frozen weights, real-time protocol; paired intervals over matched scenes.

Figure S1. **Robustness to pose noise.** SR of SE(3)-WAM under synthetic perturbations of the target pose, relative to clean poses.

C.6. Forecast Diagnostics

Table S11 compares the learned SE(3) forecast of the final model offline with constant-velocity and constant-acceleration extrapolation from the same pose history.

Table S11. **Offline prediction errors.** Target-position error at 400 ms (mm) on the same held-out observations. CV and CA are analytic extrapolations from causal pose history; the learned head reads frozen full-model states.

	Const. vel.	Harmonic	Roll/slide	Impact
CV extrapolation	—	—	—	—
CA extrapolation	—	—	—	—
Learned pose head	—	—	—	—

D. Real-Robot Experiments

D.1. Setup and Tasks

The robot is a Franka arm ([model]) with [gripper model] and a fixed RealSense D435 RGB-D camera, commanded through [a timestamped Cartesian interface] at [control frequency]. The camera records RGB-D at [resolution] and [frame rate], and we calibrate its extrinsics by [procedure] before each session. SE(3)-WAM uses this fixed third-person view; baselines whose recipes require a wrist view receive [per-method camera configuration]. All policies run on one RTX 5090. Fig. S2 shows the layout, and Table S12 defines the six tasks. T1 and T2 share a belt conveyor, T3 uses a turntable, T4 a remotely driven toy car, T5 a cue-driven collision between two balls, and T6 a ramp. Touching or pushing the target without completing the task, a single-finger hold, and an action after the target has left the workspace count as failures; [equipment-invalid trials and safety interventions are repeated or counted as failures by the following rule].

D.2. Pose Tracking and Demonstration Labels

Tracking is marker-free. Grounding DINO [7] localizes the object named by the task in the first frame, SAM 2.1 [10] gives its initial mask, and FoundationPose [14] initializes the object's 6D pose from this mask and then tracks it. [FoundationPose variant and mode (model-based with an object mesh or model-free with reference views), how the object frame is kept consistent across demonstrations and trials, and re-initialization after a loss.] The tracker runs on the control clock: at every control-period boundary, it updates the pose on the newest camera frame captured at or before the boundary, so the pose history has the 50 ms spacing used in training, and each estimate keeps the capture time of its frame. A query waits for the update of its own boundary and then runs the policy on the same GPU. While a query spans further boundaries, the tracker keeps updating at a higher GPU priority and processes only the newest frame, so that an update never queues behind older ones; the tracking time and this contention are part of the measured latency (Table S9). [Camera frame rate and the measured time per update.] When the target is lost, the estimate is marked invalid, and the previous pose is not carried forward. [How the target poses of the demonstrations, used as inputs and training labels, are obtained.] For the ball tasks, the target-state tokens keep the translation, velocity, and acceleration of the ball center and mask its orientation and angular motion, and the pose head is supervised on translation only. Table S13

Figure placeholder: Fig. S2 — real-robot setup

(a) Left, photo of the workcell from behind the robot: the third-person RGB-D camera, the robot, and the conveyor, turntable, car track, ball-collision track, and ramp at their operating positions, each labeled; the wrist camera if a baseline uses one. **(b) Right, top-down schematic to scale:** the robot base frame with its axes, the camera position and viewing direction with its distance to the workspace, the footprint and motion direction of each device for T1–T6, the grasp or interception region of each task, and the workspace bounds in cm. **Rules:** real photo, unretouched; schematic from the calibrated extrinsics and measured device positions; no repetition of the task photos of Fig. 6.

Figure S2. **Real-robot setup.** (a) The workcell. (b) Top-down layout of the camera, the robot base frame, and the devices of T1–T6 [with the workspace dimensions].

Table S12. Real-robot task definitions. $N = 100$: training demonstrations per task; 20 additional demonstrations per task are reserved for validation.

Task	Device, motion, and objects	Operating range; N	Success and failure criteria
T1	Conveyor transports a graspable cube.	[belt speed]; 100	[stable grasp and lift; failures]
T2	Conveyor transports an open box; the arm places a cube into it.	[belt speed]; 100	[placement and retention; failures]
T3	Turntable carries a cube around its axis.	[angular speed, radius]; 100	[stable grasp and lift; failures]
T4	Remotely driven toy car moves on the tabletop.	[speed, path]; 100	[car grasp and lift; failures]
T5	A cue-driven ball strikes a second ball; the arm intercepts the second ball after the collision.	[impact and outgoing speed]; 100	[post-impact capture; failures]
T6	A ball rolls or a block slides down a ramp toward the interception region.	[slope, release, speed]; 100	[capture and retention; failures]

Table S13. **Real-robot pose tracking.** Median/p95 translation (mm) and rotation (°) error of the tracker against a marker-free reference, the uncertainty of that reference, and the fraction of lost observations. Ball rotation is not used.

Target	mm	°	Ref. uncertainty	Lost
Real rigid object	—	—	—	—
Real ball	—	–	—	—

reports the tracking accuracy.

D.3. Training and Trial Protocol

We compare SE(3)-WAM, $\pi_{0.5}$, DynamicWAM, and Diffusion Policy [1], with 20 trials per task and method. All methods use the same pre-generated trial keys, which fix the object instance, the initial pose, and the motion parameters, sampled by [sampling protocol]. Trials run in [randomized method order], so that drift of the apparatus does not align with the methods. An operator resets the scene. A trial starts at the first usable observation, the robot may move at any time, and a trial ends at success, failure, or a timeout of [X] s. Demonstrations, development trials, and test trials come from separate sessions. SE(3)-WAM is fine-tuned per task from the DROID base (Appendix A.6), and the baselines are trained or fine-tuned on the same demonstrations with [per-method initialization and training settings]. For every method, we log the end-to-end time-to-action of each query on the same RTX 5090, including tracking for SE(3)-WAM; Table 4 reports its p95.

D.4. Detailed Results and Failures

Table S14 reports successes per task, 95% binomial intervals for the pooled rates, and paired differences from a bootstrap over trial keys. [The most frequent failures of SE(3)-WAM and their causes.]

Table S14. **Real-robot successes** per task (successes/trials, 20 trials per task; tasks in Table S12). All: pooled success rate (%) with its 95% binomial interval. Δ: paired difference from SE(3)-WAM (points) with a bootstrap interval over trial keys.

Method	T1	T2	T3	T4	T5	T6	All	Δ
$\pi_{0.5}$	—	—	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—	—	—
Diffusion Policy	—	—	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—	—	–

E. Preliminary Design Studies

We screened the encoder, the history representation, the forecast targets, and the coupling of world and action streams with smaller models: 4 layers of width 768, trained for 4 epochs (40,104 updates at a global batch size of 16) and evaluated with the synchronous executor that preceded our real-time protocol. Each group keeps its own encoder and seed count, stated below, and conditions are compared within a group. These studies selected the components of the final model; the five controls of Sec. 4.3 then test its central components at full scale and under the real-time protocol (Appendix C.4).

E.1. Encoder, Backbone, and History

Encoder and backbone. We compared a CNN trained with the policy, frozen DINOv3 ViT-S/16 and ViT-L/16 [11], and MAE ViT-B/16 [4] (three seeds for the MAE reference; Table S15), and selected MAE. We also compared a backbone with shared attention and stream-specific feed-forward networks against one with fully separate world and action parameters, at about 47.3M backbone parameters and with three seeds each ([source of the backbone comparison]), and selected the shared attention.

History. With a frozen DINO-L encoder and one seed per condition, we compared the current state alone, the target-motion history, RGB windows, Farnebäck flow [3], flow with motion statistics [8], codec motion vectors, video-VAE latents, and compressed windows (Table S16). Its last row replaces the past target poses of the trained history model by its current pose at test time.

E.2. Forecast Targets and Auxiliary Supervision

With a frozen DINO-L encoder and one seed per condition, we compared fifteen future targets, from target-only and image-space forecasts to role-aware combinations (Table S17). A second group with the MAE encoder and one seed removed each future appearance target from the selected configuration. In the first group, no future target improved clearly on the variant without future supervision (P00, 62.78% against 62.38% for scene flow and SE(3)); single seeds at this scale do not resolve such differences, and the full-scale ablations of Sec. 4.3 test future supervision in the final model.

Table S15. **Observation encoders** at the screening scale (Appendix E.1). Params: encoder parameters; p95: time-to-action.

Encoder	Params	Trained	SR	SR_{150}	p95
CNN	—	yes	—	—	—
DINOv3 ViT-S/16	22M	frozen	—	—	—
DINOv3 ViT-L/16	300M	frozen	—	—	—
MAE ViT-B/16*	86M	frozen	—	—	—

Table S16. **History representations** at the screening scale (frozen DINO-L encoder, one training seed, 1,400 scenes). The first eight conditions receive the current target pose; the ninth adds the target-motion history. Last row: the ninth model with its past target poses replaced by the current pose at test time and the kinematic features recomputed.

History input	Success / 1,400	SR (%)
Current image	748	53.43
RGB window	662	47.29
Optical flow	720	51.43
Flow + motion statistics	723	51.64
Codec motion vectors	733	52.36
Video-VAE latent	710	50.71
Compressed window (age)	634	45.29
Compressed window (uniform)	670	47.86
Current image + target-motion history	824	58.86
stationary history at test time	9	0.64

E.3. World–Action Coupling

With the MAE encoder and three seeds per condition, we compared six couplings of the world and action streams. R0 disables every forecast loss. R1 adds one object query and one query per patch to the world stream, which attend to $\mathcal{S}$, $\mathcal{M}$, and each other and carry the forecast readouts, without any edge to the actions. R2 lets the action tokens attend to these queries at every layer, R3 lets the queries attend to the actions, and R4 opens both edges; in R3 and R4, the world states are recomputed at every Euler step. R5 decodes the forecast once per query and encodes the target trajectory and the flow with visibility, with rotations in a continuous 6D form, through a geometry encoder (0.37M parameters) into tokens that the actions attend to. Future supervision without

Table S17. **Forecast targets** at the screening scale. The first group uses a frozen DINO-L encoder and seed 0, with its 140 development scenes and the 1,260 remaining scenes reported separately. The second group uses the MAE encoder and seed 1 on 1,400 scenes; it keeps scene flow, SE(3), and the current-target RGB reconstruction and removes future appearance targets. Compare conditions within each group. Targets are cumulative from the current state at the eight horizons, except codec motion: SE(3), the target pose; 3D scene flow, base-frame displacements of the 14×14 patch centers; 2D tracks and dense 2D displacement, image displacements of target points, grid points, or all pixels; codec motion, the motion vectors of a video codec; video-VAE latent, latents of a frozen Wan2.2 VAE; object DINO, mask-pooled DINOv3 features of the target.

ID	Future supervision	Dev. success / 140	Success / 1,260	SR_{1260} (%)
P00	None	88	791	62.78
P01	3D scene flow	95	782	62.06
P02	SE(3)	92	789	62.62
P03	Scene flow + SE(3)	93	786	62.38
P04	Video-VAE latent + SE(3)	92	783	62.14
P05	Dense 2D displacement + SE(3)	87	746	59.21
P06	Codec motion + SE(3)	86	795	63.10
P07	Sparse scene 2D tracks + SE(3)	90	807	64.05
P08	Scene flow + object DINO	89	794	63.02
P09	Scene flow + target 2D tracks	88	808	64.13
A_α	P03 + motion magnitude	89	778	61.75
P12	P03 + depth	90	791	62.78
P13	P03 + DINO	92	788	62.54
P14	P03 + dynamic RGB	89	790	62.70
P15	P03 + depth / DINO / dynamic RGB	94	802	63.65
AUX0 MAE-B: future appearance auxiliaries		Success / 1,400		SR_{1400} (%)
None		894		63.86
Depth only		867		61.93
DINO only		946		67.57
Dynamic RGB only		926		66.14
All three (existing reference)		995		71.07

Table S18. **Couplings R0–R5** at the screening scale (MAE encoder, three training seeds each, 1,400 development scenes per seed; Appendix E.3). SR: mean over seeds (%); s.d.: standard deviation over seeds (points).

Coupling	SR	s.d.
R0 (future losses off)	60.45	3.57
R1 (queries, no edge)	63.60	2.34
R2 (world→action)	62.52	1.97
R3 (action→world)	64.21	2.13
R4 (bidirectional)	63.00	4.19
R5 (decoded geometry)	63.95	1.91

any edge to the actions (R1) exceeded R0 by 3.14 points on average, with seed-wise differences of −1.29, +9.64, and +1.07. Feeding the forecasts to the actions gave no consistent further gain: R2 and R5 differed from R1 by −1.07 and +0.36 points on average, with standard deviations of 4.28 and 3.45 points over the seed-wise differences, and the effects of both cross-stream edges changed sign across seeds (Table S18). We therefore use the forecasts only as training supervision, which also keeps deployment free of their cost.

References

[1] Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin C. M. Burchfiel, and Shuran Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems*, 2023. 7, 11

[2] Heng Fang, Shangru Li, Shuhan Wang, Xuanyang Xi, Dingkang Liang, and Xiang Bai. Towards generalizable robotic manipulation in dynamic environments. *arXiv preprint arXiv:2603.15620*, 2026. 7, 9

[3] Gunnar Farnebäck. Two-frame motion estimation based on polynomial expansion. In *Proceedings of the Scandinavian Conference on Image Analysis*, pages 363–370, 2003. 12

[4] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16000–16009, 2022. 1, 12

[5] Wenlong Huang, Yu-Wei Chao, Arsalan Mousavian, Ming-Yu Liu, Dieter Fox, Kaichun Mo, and Li Fei-Fei. PointWorld: Scaling 3D world models for in-the-wild robotic manipulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026. 2

[6] Alexander Khazatsky, Karl Pertsch, Suraj Nair, et al. DROID: A large-scale in-the-wild robot manipulation dataset. In *Proceedings of Robotics: Science and Systems*, 2024. 2

[7] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. In *European Conference on Computer Vision*, 2024. 10

[8] Yunfan Lou, Hewen Gao, Xiyu Zhu, Zhuoran Qiao, Xuan Han, Yifan Yang, Yifan Ye, Boxian Yao, and Zhibo Pang. DynamicWAM: Dual-path motion conditioning for world-action models in dynamic manipulation. *arXiv preprint arXiv:2608.00793*, 2026. 7, 12

[9] Physical Intelligence, Kevin Black, Noah Brown, James Darpinian, et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025. 7

[10] Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. In *International Conference on Learning Representations*, 2025. 10

[11] Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, et al. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025. 1, 2, 12

[12] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 5026–5033, 2012. 3

[13] Yuran Wang, Siqiao Huang, Mingleyang Li, Chenhao Zhang, Jiaqi Liang, Weiyang Jin, Yue Chen, Xuemin Chi, Donghao Zhou, Qize Yu, Yu-Kai Wang, Yuhan Rui, Shenzhe Yao, Zhen Yuan, Zhenhao Shen, Kefei Zhu, Zijie Zhu, Ning Gao, Xiaowei Chi, Guanqi He, Shanghang Zhang, Hao Dong, Lin Shao, and Hang Zhao. OpenWAM: An open, modular exploration towards systematic world-action model pretraining. *arXiv preprint arXiv:2609.07398*, 2026. 7

[14] Bowen Wen, Wei Yang, Jan Kautz, and Stan Birchfield. FoundationPose: Unified 6D pose estimation and tracking of novel objects. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17868–17879, 2024. 10

[15] Haozhe Xie, Beichen Wen, Jiarui Zheng, Zhaoxi Chen, Fangzhou Hong, Haiwen Diao, and Ziwei Liu. DynamicVLA: A vision-language-action model for dynamic object manipulation. *arXiv preprint arXiv:2601.22153*, 2026. 7

[16] Wenyao Zhang, Hongsi Liu, Zekun Qi, Yunnan Wang, Xin-Qiang Yu, Jiazhao Zhang, Runpei Dong, Jiawei He, He Wang, Zhizheng Zhang, Li Yi, Wenjun Zeng, and Xin Jin. DreamVLA: A vision-language-action model dreamed with comprehensive world knowledge. In *Advances in Neural Information Processing Systems*, pages 24195–24228, 2025. 7

[17] Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware. In *Proceedings of Robotics: Science and Systems*, 2023. 7

[18] Yuke Zhu, Josiah Wong, Ajay Mandlekar, Roberto Martín-Martín, Abhishek Joshi, Kevin Lin, Abhiram Maddukuri, Soroush Nasiriany, and Yifeng Zhu. robosuite: A modular simulation framework and benchmark for robot learning. *arXiv preprint arXiv:2009.12293*, 2020. 3